

QUESTION ONE (15 marks) Use a separate writing booklet.**Marks**

(a) Find $\int_0^4 \frac{1}{\sqrt{2x+1}} dx$. **2**

(b) Find $\int \tan^3 x \sec^2 x dx$. **2**

(c) Find $\int \frac{x}{x^2 - 4x + 8} dx$. **3**

(d) (i) Find the values of A and B such that $\frac{3x^2 - 10}{x^2 - 4x + 4} = 3 + \frac{A}{x-2} + \frac{B}{(x-2)^2}$. **2**

(ii) Find $\int \frac{3x^2 - 10}{x^2 - 4x + 4} dx$. **2**

(e) Use integration by parts twice, to show that $\int_1^e \sin(\ln x) dx = \frac{e}{2}(\sin 1 - \cos 1) + \frac{1}{2}$. **4**

QUESTION TWO (15 marks) Use a separate writing booklet.**Marks**

(a) Simplify $|\cos \theta + i \sin \theta|$. **1**

(b) Express $\frac{i^5(1-i)}{2+i}$ in the form $a + ib$ where a and b are rational. **2**

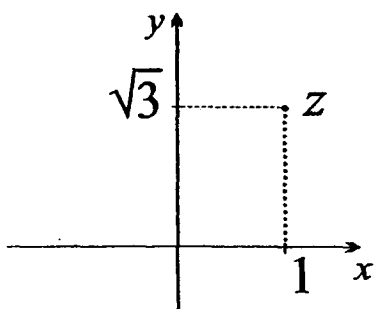
(c) By drawing a diagram, or otherwise, find the solutions of $z^5 = -1$. **2**

(d) Graph the region in the Argand diagram which simultaneously satisfies **3**

$$1 \leq |z - i| \leq 2 \text{ and } \operatorname{Im} z \geq 0.$$

(e) Find the complex number ϕ if $1 + i$ is a root of the equation $z^2 + \phi z - i = 0$. **2**

(f)



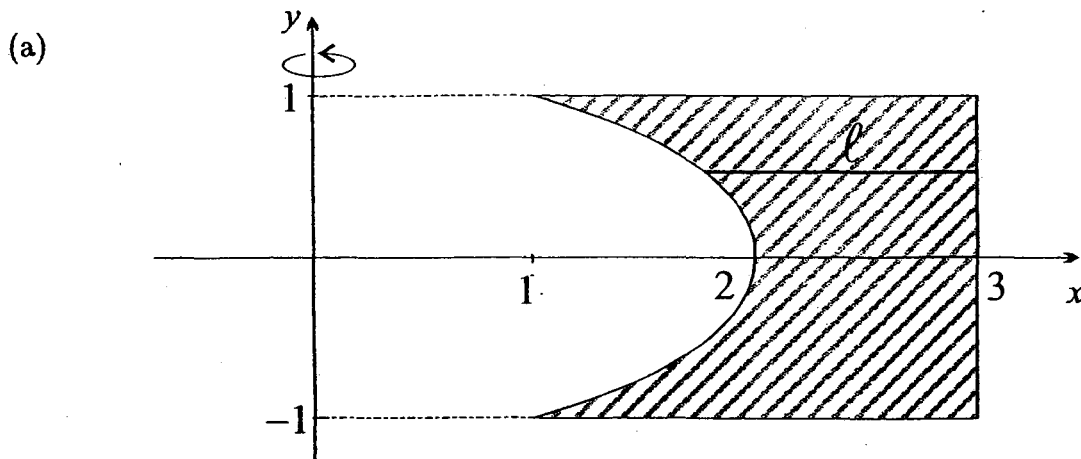
Suppose that $z = 1 + \sqrt{3}i$ and $\omega = (\operatorname{cis} \alpha)z$ where $-\pi < \alpha \leq \pi$.

(i) Find the argument of z . **1**

(ii) Find the value of α if ω is purely imaginary and $\operatorname{Im}(\omega) > 0$. **2**

(iii) Find the value of $\arg(z + \omega)$ if ω is purely imaginary and $\operatorname{Im}(\omega) > 0$. **2**

Exam continues next page ...

QUESTION THREE (15 marks) Use a separate writing booklet.**Marks**

The diagram above shows the region bounded by the curve $x = 2 - y^2$ and the lines $x = 3$, $y = 1$ and $y = -1$. This region is rotated about the y -axis to form a solid. The interval ℓ at height y sweeps out an annulus.

- (i) Show that the annulus at height y has area equal to

2

$$\pi(5 + 4y^2 - y^4).$$

- (ii) Find the volume of the solid.

2

- (b) Consider the function $f(x) = \frac{1}{1+x^3}$.

- (i) Show that there is a horizontal point of inflexion at $x = 0$.

2

- (ii) Find the vertical asymptote and the horizontal asymptote.

2

- (iii) Sketch $y = f(x)$ showing the features from parts (a) and (b) and the y -intercept.

2

- (iv) On a separate diagram sketch $y = |f(x)|$.

1

- (v) On a separate diagram sketch $y^2 = f(x)$.

2

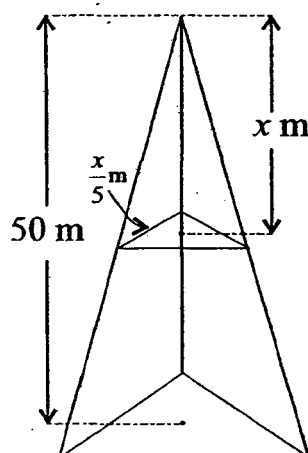
- (vi) On a separate diagram sketch $y = e^{f(x)}$.

2

Exam continues overleaf ...

QUESTION FOUR (15 marks) Use a separate writing booklet.**Marks**(a) Consider the polynomial equation $x^3 - 3x^2 + x - 5 = 0$ which has roots α , β and γ .(i) Show that $\alpha + \beta = 3 - \gamma$.**1**(ii) Write down similar expressions for $\alpha + \gamma$ and $\beta + \gamma$ and hence find a polynomial equation which has the roots $\alpha + \beta$, $\alpha + \gamma$ and $\beta + \gamma$.**2**

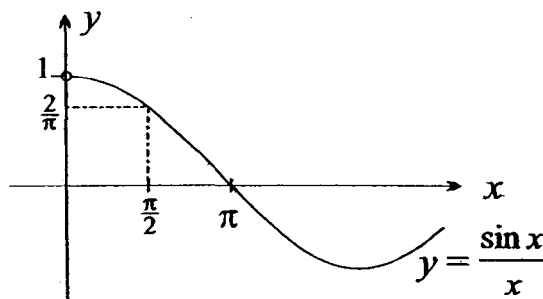
(b)



The diagram above shows a monument 50 metres high. A horizontal cross section x metres from the top is an equilateral triangle with sides $\frac{x}{5}$ metres. Use integration to find the volume of the monument.

3

(c)



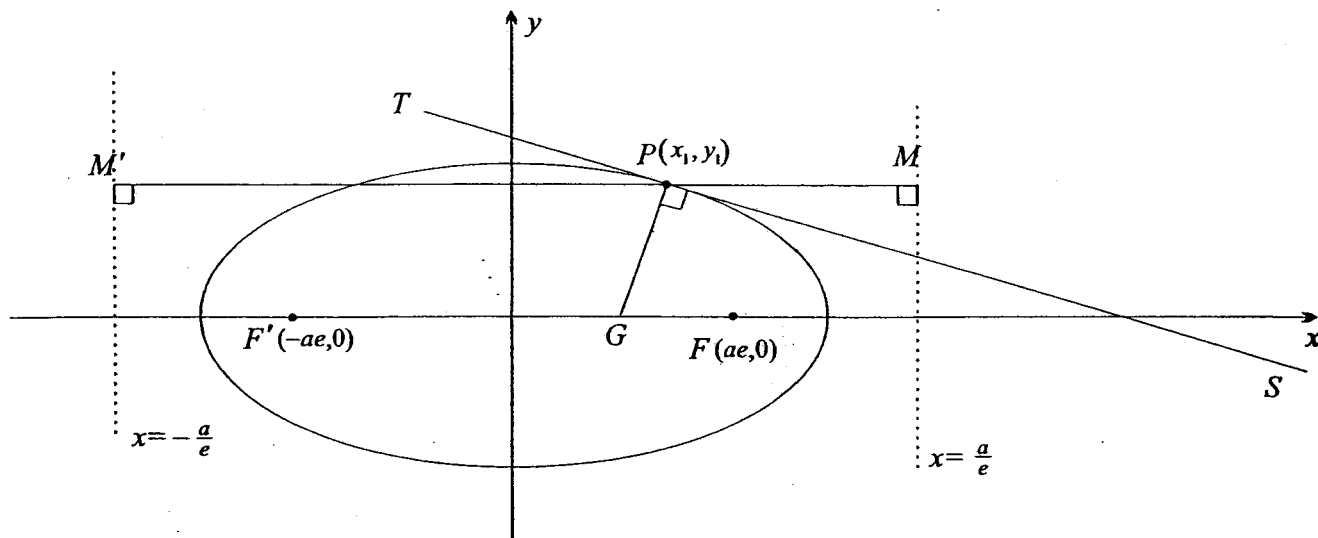
Use the method of cylindrical shells to find the volume of the solid formed when the region bounded by the curve $y = \frac{\sin x}{x}$ and the lines $y = 0$ and $x = \frac{\pi}{2}$ is rotated about the y -axis.

4(d) An hyperbola is defined parametrically by $x = 3 \sec \theta$ and $y = 4 \tan \theta$.(i) Write the equation of the curve in Cartesian form and show that the eccentricity is $\frac{5}{3}$.**2**(ii) Sketch the curve showing its x -intercepts, foci, directrices and asymptotes.**4**

Exam continues next page ...

QUESTION FIVE (15 marks) Use a separate writing booklet.**Marks**

(a)



The diagram above shows the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with foci $F(ae, 0)$ and $F'(-ae, 0)$.

$P(x_1, y_1)$ is any point on the ellipse.

Let M and M' be the feet of the perpendiculars from P to the directrices $x = \frac{a}{e}$ and $x = -\frac{a}{e}$.

Line TS is a tangent to the ellipse at P and G is the point where the normal at P meets the x -axis.

(i) Show that the equation of the normal at P is $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$.

3

(ii) Show that the point G has co-ordinates $(e^2x_1, 0)$.

3

(iii) Show that the distance PF is $a - ex_1$.

2

(iv) Show that $\frac{PF}{FG} = \frac{PF'}{F'G}$.

2

(b) (i) Show that $1 - \cos 2\theta - i \sin 2\theta = 2 \sin \theta (\sin \theta - i \cos \theta)$.

2

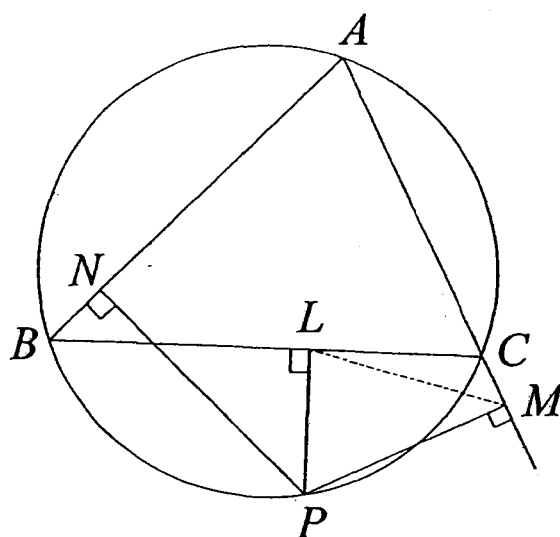
(ii) Given that $\frac{z-1}{z} = \text{cis } \frac{2\pi}{5}$, show that $z = \frac{1}{2}(1 + i \cot \frac{\pi}{5})$.

3

Exam continues overleaf ...

QUESTION SIX (15 marks) Use a separate writing booklet.**Marks**

(a)



In the diagram above, ABC is a triangle with the circumcircle through points A , B and C drawn. P is another point on the minor arc BC . Points L , M and N are the feet of the perpendiculars from P to the sides BC , CA and AB respectively.

(i) Copy the diagram and explain why P , L , N and B are concyclic. 1

(ii) Explain why P , L , C and M are concyclic. 1

(iii) Let $\angle PLM = \alpha$.

(α) Show that $\angle ABP = \alpha$. 2

(β) Hence show that M , L and N are collinear. 2

- (b) A particle of unit mass is thrown vertically downwards with an initial velocity of v_0 . It experiences a resistive force of magnitude kv^2 where v is its velocity. Taking downwards as the positive direction, the equation of motion of the particle is given by

$$\ddot{x} = g - kv^2.$$

Let V be the terminal velocity of the particle.

(i) Explain why $V = \sqrt{\frac{g}{k}}$. 1

(ii) Show that $v^2 = V^2 + (v_0^2 - V^2)e^{-2kx}$. 4

- (c) Let $z = x + iy$ be any non-zero complex number such that $z + \frac{1}{z} = k$, where k is a real number.

(i) Prove that either $y = 0$ or $x^2 + y^2 = 1$. 2

(ii) Show that if $y = 0$ then $|k| \geq 2$. 2

Exam continues next page ...

QUESTION SEVEN (15 marks) Use a separate writing booklet.**Marks**(a) (i) Write down $\cos 2\theta$ in terms of $\tan \theta$.**1**(ii) Show that $\cos 4\theta = \frac{1 - 6 \tan^2 \theta + \tan^4 \theta}{1 + 2 \tan^2 \theta + \tan^4 \theta}$.**3**(iii) Deduce that $\tan^2 \frac{\pi}{8} + \tan^2 \frac{3\pi}{8} = 6$.**3**(b) Consider the equation $z^7 = 1$.This equation has seven roots $1, \rho, \rho^2, \dots, \rho^6$, where $\rho = \text{cis } \frac{2\pi}{7}$.Let $\alpha = \rho + \rho^2 + \rho^4$ and $\theta = \rho^3 + \rho^5 + \rho^6$.(i) Express ρ^9 as a lower positive power of ρ .**1**(ii) Simplify $\alpha + \theta$.**2**(iii) Simplify $\alpha\theta$.**2**(iv) Form a quadratic equation with α and θ as roots.**1**(v) Deduce that $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} = -\frac{1}{2}$.**2****Exam continues overleaf ...**

QUESTION EIGHT (15 marks) Use a separate writing booklet.**Marks**

- (a) The sequence $a_1, a_2, \dots, a_n$ is defined by $a_n = \frac{(2n)!}{2^n n!}$.

4

Show by induction on n that a_n is an odd positive integer.

- (b) Suppose that $y = f(x)$ is an increasing function for $x \geq 1$.
Suppose also that $f(x) \geq 0$ for $x \geq 1$.

- (i) Explain, with the aid of a diagram, why

2

$$f(1) + f(2) + \dots + f(n-1) < \int_1^n f(x) dx < f(2) + f(3) + \dots + f(n).$$

- (ii) Show that $\int_1^n \ln x dx = n \ln n - n + 1$.

2

- (iii) Use parts (i) and (ii) to deduce that, for $n > 1$:

(α) $n! > \frac{n^n}{e^{n-1}}$

3

(β) $n! < \frac{n^{n+1}}{e^{n-1}}$

2

- (iv) Find $\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n}$. (You may assume that $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.)

2**END OF EXAMINATION**

Question One

(1)

$$\begin{aligned}
 (a) \quad & \int_0^4 (2x+1)^{\frac{1}{2}} dx \\
 &= \left[\frac{(2x+1)^{\frac{1}{2}}}{2 \times \frac{1}{2}} \right]_0^4 \quad \checkmark \\
 &= \sqrt{9} - \sqrt{1} \quad \checkmark \\
 &= 2.
 \end{aligned}$$

$$(b) \quad I = \int \tan^3 x \sec^2 x \, dx$$

$$\begin{aligned}
 \text{Let } u &= \tan x \\
 du &= \sec^2 x \, dx \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 I &= \int u^3 \, du \\
 &= \frac{u^4}{4} + C \\
 &= \frac{\tan^4 x}{4} + C \quad \checkmark
 \end{aligned}$$

$$(c) \quad \int \frac{x}{x^2 - 4x + 8} \, dx$$

$$= \frac{1}{2} \int \frac{2x - 4 + 4}{x^2 - 4x + 8} \, dx \quad \checkmark$$

$$= \frac{1}{2} \int \frac{2x - 4}{x^2 - 4x + 8} \, dx + \int \frac{2}{(x-2)^2 + 4} \, dx$$

$$= \frac{1}{2} \ln(x^2 - 4x + 8) + 2 \times \frac{1}{2} \tan^{-1} \frac{x-2}{2} + C$$

$$\begin{aligned}
 &= \frac{1}{2} \ln(x^2 - 4x + 8) + \tan^{-1} \frac{x-2}{2} + C \\
 &\quad \checkmark \qquad \qquad \checkmark
 \end{aligned}$$

(2)

$$(d) \quad (1) \quad \frac{3x^2 - 10}{x^2 - 4x + 4} = 3 + \frac{A}{x-2} + \frac{B}{(x-2)^2}$$

$$3x^2 - 10 = 3(x-2)^2 + A(x-2) + B$$

$$\text{subst } x=2 : \quad 12 - 10 = B \quad \checkmark$$

$$B = 2$$

$$\text{subst } x=0 : \quad -10 = 12 + B - 2A$$

$$-22 = 2 - 2A \quad \checkmark$$

$$-24 = -2A \quad \checkmark$$

$$A = 12$$

$$\int \frac{3x^2 - 10}{x^2 - 4x + 4} dx = \int 3 + \frac{12}{x-2} + \frac{2}{(x-2)^2}$$

$$= 3x + 12 \ln|x-2| - \frac{4}{x-2} + C \quad \checkmark$$

$$(e) \quad \int_1^e \sin(\ln x) \frac{d}{dx}(x) dx$$

$$= x \sin(\ln x) \Big|_1^e - \int_1^e x \cdot \cos(\ln x) \cdot \frac{1}{x} dx \quad \checkmark$$

$$= x \sin(\ln x) \Big|_1^e - \int_1^e \cos \ln x \frac{d}{dx}(x) dx$$

$$= e \sin 1 - 0 - \left\{ [x \cos(\ln x)]_1^e - \int_1^e x \sin(\ln x) \cdot \frac{1}{x} dx \right\} \quad \checkmark$$

$$= e \sin 1 - \left\{ e \cos 1 - 1 + \int_1^e \sin(\ln x) dx \right\}$$

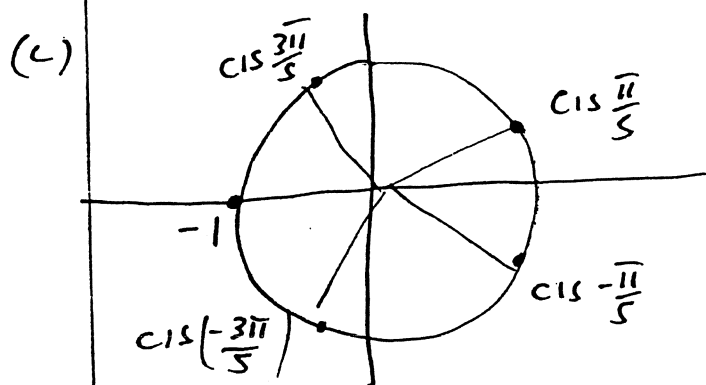
$$\text{So } 2 \int_1^e \sin(\ln x) dx = e \sin 1 - e \cos 1 + 1 \quad \checkmark$$

$$\int_1^e \sin(\ln x) dx = \frac{e}{2} [\sin 1 - \cos 1] + \frac{1}{2} \quad \checkmark$$

Question Two

(a) $|\cos \theta + i \sin \theta| = 1$ ✓

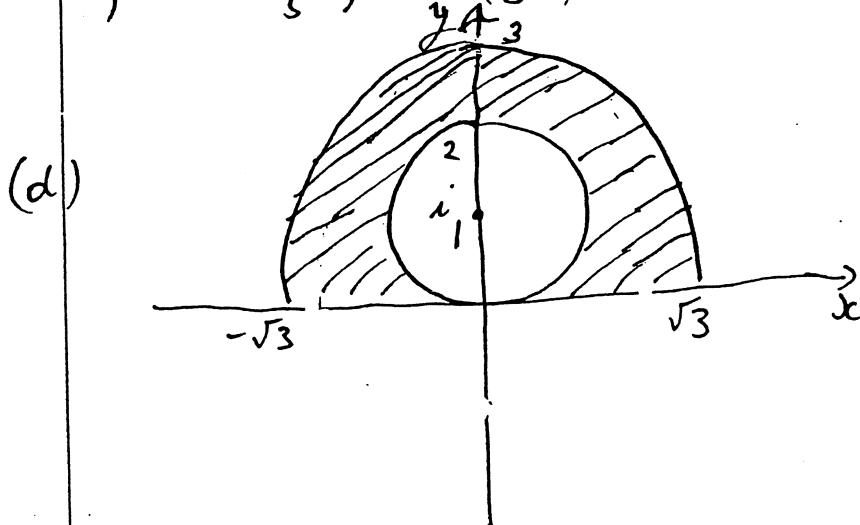
(b)
$$\begin{aligned} \frac{z^5(1-z)}{2+i} &= \frac{z(1-z)}{2+i} \\ &= \frac{z+1}{2+i} \\ &= \frac{(1+i)(2-z)}{4+1} \quad \checkmark \\ &= \frac{2-z+2i+1}{5} \\ &= \frac{3}{5} + \frac{1}{5}i \quad \checkmark \end{aligned}$$



Clearly $z = -1$ is a root.
The other four roots are equally spaced around the circle.
✓

Solutions to $z^5 = -1$ are

$-1, \text{cis } \frac{\pi}{5}, \text{cis}(-\frac{\pi}{5}), \text{cis } \frac{3\pi}{5} \text{ and } \text{cis}(-\frac{3\pi}{5})$ ✓



✓ annulus (0,1) centre
✓ above x axis
✓ correct shading (radius between 1 and 2)

(e) $1+i$ is a root

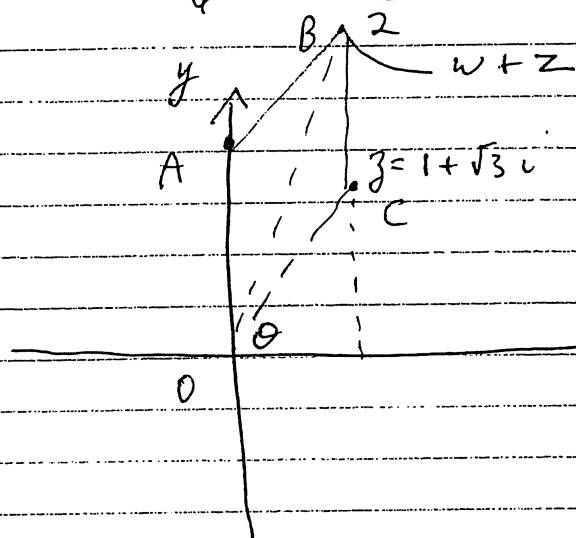
$$\therefore (1+i)^2 + \phi(1+i) - i = 0 \quad \checkmark$$

$$1+2i-1 + \phi(1+i) - i = 0$$

$$\phi = \frac{-i}{1+i} \times \frac{1-i}{1-i}$$

$$\phi = \frac{-1-i}{2} \quad \checkmark$$

(f)



(i) $\tan \theta = \sqrt{3}$
 $\theta = \frac{\pi}{3}$

$\arg(z) = \frac{\pi}{3} \quad \checkmark$

(ii) Multiplication by $\text{cis } \alpha$ is a rotation anticlockwise through α $\alpha + \frac{\pi}{3} = \frac{\pi}{2} \quad \checkmark$

So if w is purely imaginary and $\text{Im}(z) > 0$
 then $\alpha = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6} \quad \checkmark$

(iii) $OABC$ is a rhombus
 Diagonals bisect vertex $\checkmark$

So $\arg(z+w) = \frac{\pi}{2} + \frac{\pi}{12}$
 $= \frac{5\pi}{12} \quad \checkmark$

(5)

$$\begin{aligned}
 (a) \quad (i) \quad A(y) &= \pi(r_1^2 - r_2^2) \\
 &= \pi(3^2 - (2 - y^2)^2) \quad \checkmark \\
 &= \pi(9 - 4 + 4y^2 - y^4) \\
 &= \pi(5 + 4y^2 - y^4) \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad V &= 2\pi \int_0^1 (5 + 4y^2 - y^4) dy \quad \checkmark \\
 &= 2\pi \left[5y + \frac{4y^3}{3} - \frac{y^5}{5} \right]_0^1 \\
 &= 2\pi \left[5 + \frac{4}{3} - \frac{1}{5} \right] \\
 &= \frac{2\pi [75 + 20 - 3]}{15} = \frac{184\pi}{15} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad (i) \quad f(x) &= (1+x^3)^{-1} \\
 f'(x) &= -\frac{3x^2}{(1+x^3)^2} \quad \checkmark
 \end{aligned}$$

Start point where $x = 0$

Horizontal point
of inflection
where $x = 0$
 $y = 1$

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
y'	-ve	0	-ve

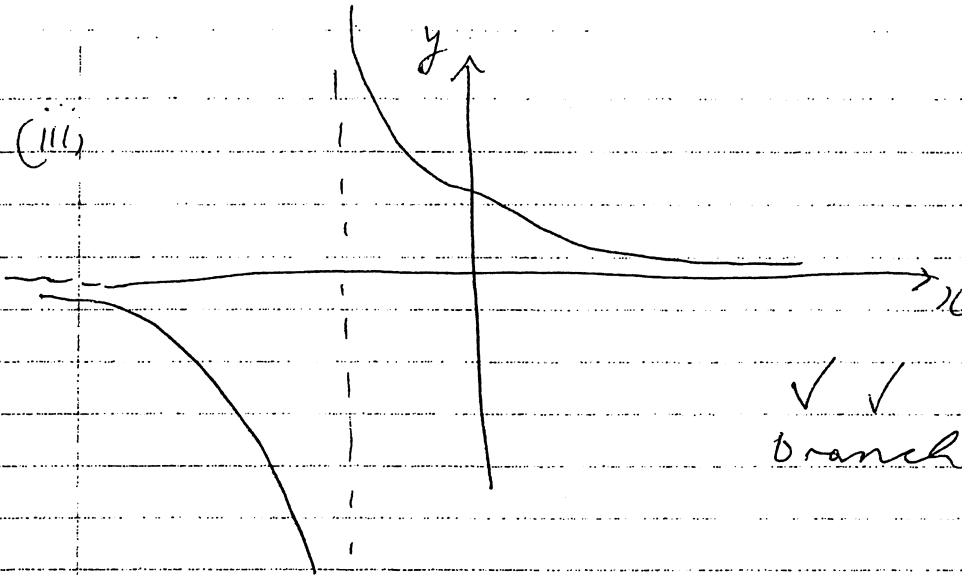
(ii) Vertical asymptote at $x = -1$ $\checkmark$

As $x \rightarrow \infty$, $y \rightarrow 0^+$
 $x \rightarrow -\infty$, $y \rightarrow 0^-$ $\checkmark$

Horizontal asymptote at $y = 0$.

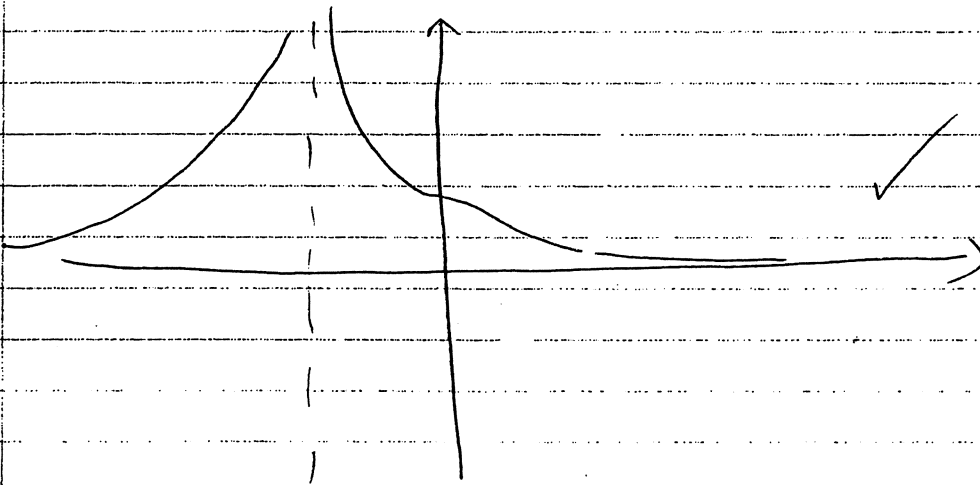
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(iii)



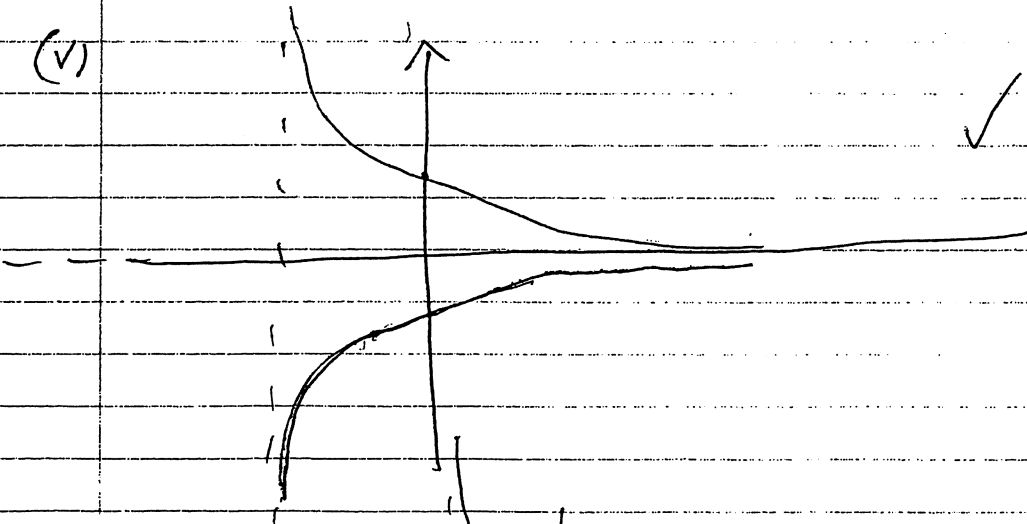
✓ ✓ for each
branch correct.

(iv)



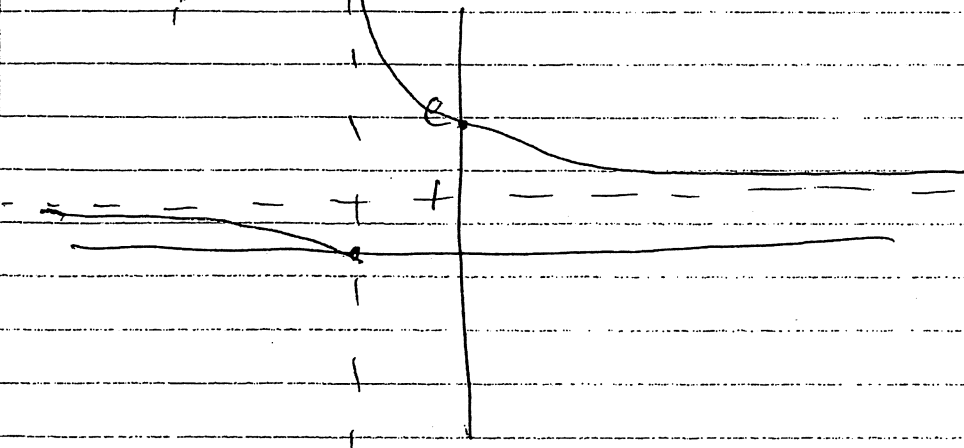
✓

(v)



correct domain
✓ ✓ reflection.

(vi)



✓ ✓

(7)

Question Four

(a) $x^3 - 3x^2 + x - 5 = 0$

(i) $\alpha + \beta + \gamma = -\frac{b}{a}$

$\alpha + \beta + \gamma = 3$ ✓

$\alpha + \beta = 3 - \gamma$

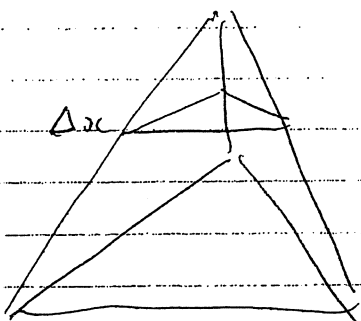
(ii) $\beta + \gamma = 3 - \alpha$ and $\alpha + \gamma = 3 - \beta$

The polynomial equation has roots
 $3 - \alpha$, $3 - \beta$, $3 - \gamma$.

Transformation is $y = 3 - x$
 $x = 3 - y$ ✓

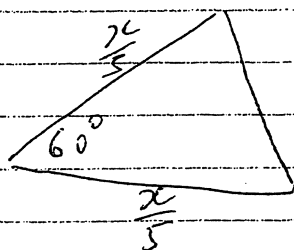
New equation is $(3 - y)^3 - 3(3 - y)^2 + (3 - y) - 5 = 0$ ✓

(b)



$V = \lim_{\Delta x \rightarrow 0} \sum A(x) \Delta x$

$V = \int_0^{50} \frac{1}{2} \left(\frac{x}{5} \right)^2 \cdot \frac{\sqrt{3}}{2} dx$ ✓



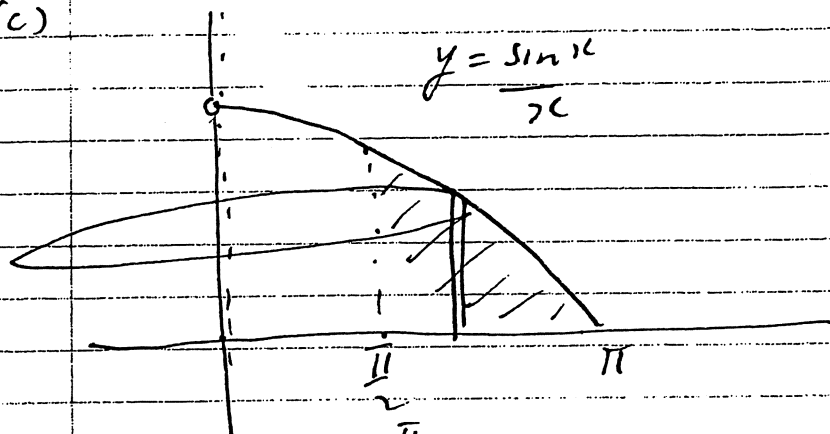
$A(x) = \frac{1}{2} \left(\frac{x}{5} \right)^2 \sin 60^\circ$
 $= \frac{x^2}{50} \cdot \frac{\sqrt{3}}{2}$ ✓

$= \frac{\sqrt{3}}{100} \left[\frac{x^3}{3} \right]_0^{50}$

$= \frac{\sqrt{3}}{100} \times \frac{50^3}{3}$

$= \frac{1250\sqrt{3}}{6} \text{ units}^3$ ✓

(c)



$$\Delta V = 2\pi x f(x) \Delta x$$

$$V = 2\pi \int_{\pi/2}^{\pi} x \cdot \frac{\sin x}{x} dx$$

$$= 2\pi \int_{\pi/2}^{\pi} \sin x dx$$

$$= 2\pi \left[-\cos x \right]_{\pi/2}^{\pi}$$

$$= 2\pi \left[-(-1) - 0 \right] = 2\pi$$

(d)

$$x = 3 \sec \theta \quad y = 4 \tan \theta$$

(i)

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$a^2 = 9, \quad b^2 = 16$$

$$b^2 = a^2(e^2 - 1)$$

$$16 = 9(e^2 - 1)$$

$$\frac{16}{9} + 1 = e^2$$

$$e^2 = \frac{25}{9}$$

$$e = \frac{5}{3}$$

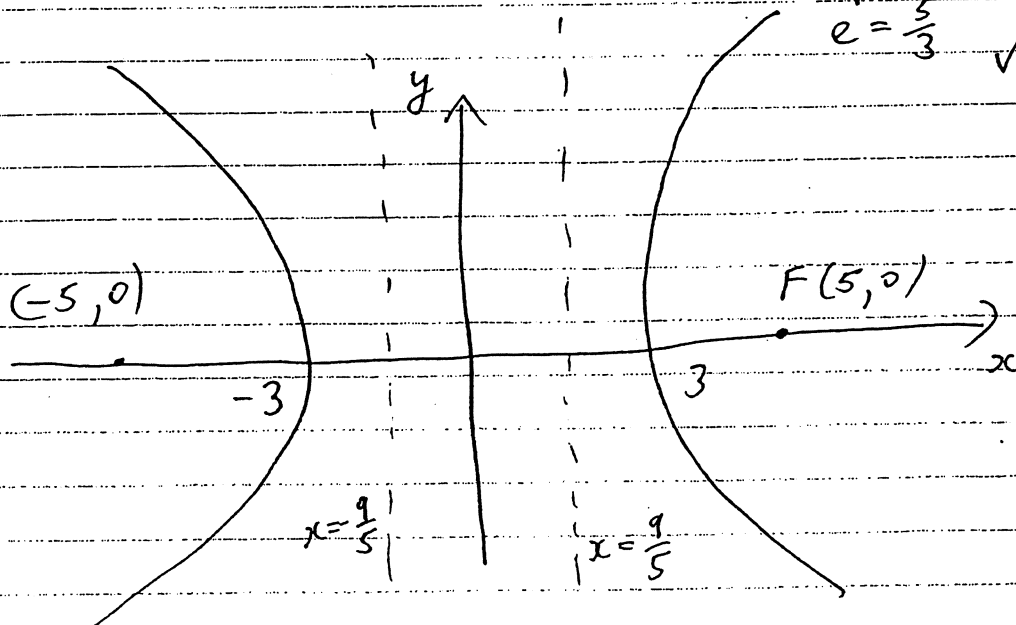
(ii)

x-ints ✓

foci ✓

directrices ✓

asymptotes ✓



Question Five

(a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(i) Differentiate implicitly wrt x

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

At $P(x_1, y_1)$, gradient $= -\frac{b^2 x_1}{a^2 y_1}$

So gradient of normal is $\frac{a^2 y_1}{b^2 x_1}$

Eqn of normal is

$$y - y_1 = \frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

$$\frac{y}{y_1} - 1 = \frac{a^2}{b^2} \left(\frac{x}{x_1} - 1 \right)$$

$$\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

(ii) When $y = 0$, $x = \frac{a^2 - b^2}{a^2} x_1$

But $\frac{a^2 - b^2}{a^2} = e^2$

So $G = (e^2 x_1, 0)$

(iii) Now $PF = e \cdot PM$ (defn of ellipse)

$$PF = e \left(\frac{a}{e} - x_1 \right)$$

$$= a - e x_1$$

$$(iv) \quad \frac{PF}{FG} = \frac{a - ex_1}{ae - e^2 x_1} = \frac{1}{e} \quad \checkmark$$

$$\frac{PF'}{F'G} = \frac{a + ex_1}{a + e^2 x_1} = \frac{1}{e} \quad \checkmark$$

$$\text{So } \frac{PF}{FG} = \frac{PF'}{F'G}$$

$$\begin{aligned} (b) \quad (i) \quad LHS &= 1 - \cos 2\alpha - i \sin 2\alpha \\ &= 1 - (1 - 2\sin^2 \alpha) - 2i \sin \alpha \cos \alpha \quad \checkmark \\ &= 2\sin^2 \alpha - 2i \sin \alpha \cos \alpha \\ &= 2\sin \alpha (\sin \alpha - i \cos \alpha) \quad \checkmark \\ &= RHS \end{aligned}$$

$$(ii) \quad \frac{z-1}{z} = \operatorname{cis} \frac{2\pi}{5}$$

$$z - 1 = z \cos \frac{2\pi}{5} + i z \sin \frac{2\pi}{5}$$

$$z \left(1 - \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5} \right) = 1$$

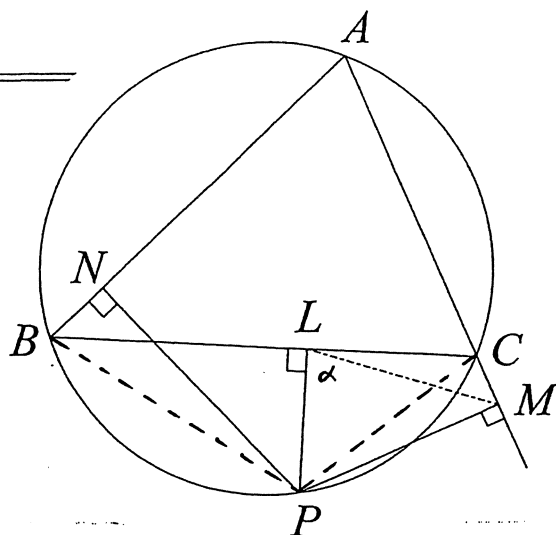
$$z \times 2 \sin \frac{\pi}{5} \left(\sin \frac{\pi}{5} - i \cos \frac{\pi}{5} \right) = 1 \quad \text{by part (i)} \quad \checkmark$$

$$2z \sin \frac{\pi}{5} = \sin \frac{\pi}{5} + i \cos \frac{\pi}{5} \quad \checkmark$$

$$z = \frac{1}{2} \left(1 + i \cot \frac{\pi}{5} \right) \quad \checkmark$$

Q 6

a)



(i) $\angle BLP = \angle BNP$ (given) ✓

So B, L, N and P are concyclic by converse of angles standing on the same arc

(ii) $\angle PLC + \angle PMC = 180^\circ$ (given)

So P, L, C and M are concyclic by converse of opposite angles of a cyclic quad ✓

(iii) (a) $\angle PCM = \angle PLM$ (angles standing on the same arc PM) ✓
 $= \alpha$

$\angle ABP = \angle PCM$ (opposite interior angle of cyclic quad ABPC) ✓
 $= \alpha$

(b) $\angle NBP = \alpha$ (same as $\angle ABP$)

So $\angle NLP = 180^\circ - \alpha$ (opp angles of cyclic quad BNLP) ✓

So $\angle NLM = \angle NLP + \angle MLP$
 $= 180^\circ - \alpha + \alpha$
 $= 180^\circ$ ✓

$\therefore$ N, L and M are collinear.

[5] (i) Terminal velocity when $ac = 0$

$g - KV^2 = 0$ ✓

$V^2 = \frac{g}{K}$

So $V = \sqrt{\frac{g}{K}}$ (since $V_0 > 0$ initially and so cannot change sign)

(b) (ii) $v \frac{dv}{dx} = g - kv^2$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$\frac{dx}{dv} = -\frac{1}{2k} \frac{-2kv}{g - kv^2} \quad \checkmark$$

$$x = -\frac{1}{2k} (g - kv^2) + c$$

When $x=0$ } $c = \frac{1}{2k} (g - kv_0^2)$
 $v=v_0$ } $\checkmark$

$$x = -\frac{1}{2k} \ln \left(\frac{g - kv^2}{g - kv_0^2} \right)$$

$$\frac{g - kv^2}{g - kv_0^2} = e^{-2kx}$$

$$g - kv^2 = (g - kv_0^2) e^{-2kx} \quad \checkmark$$

$$-kv^2 = -g + (g - kv_0^2) e^{-2kx}$$

$$v^2 = \frac{g}{k} + \left(v_0^2 - \frac{g}{k} \right) e^{-2kx}$$

$$v^2 = v_0^2 + (v_0^2 - v^2) e^{-2kx} \quad \checkmark$$

(c) (i) Let $z = x + iy$

$$z = x + iy + \frac{1}{x + iy} \times \frac{x - iy}{x^2 - iy^2} \quad \checkmark$$

$\text{Im}(z) = 0$ So $y + \frac{-y}{x^2 + y^2} = 0$

$$y \left(1 - \frac{1}{x^2 + y^2} \right) = 0 \quad \checkmark$$

$$y = 0 \text{ or } x^2 + y^2 = 1$$

(ii) $x + \frac{1}{x} = k$

$$\therefore \Delta \geq 0 \quad \checkmark$$

$$x^2 - kx + 1 = 0$$

$$k^2 - 4 \geq 0$$

has real roots.

$$|k| \geq 2 \quad \checkmark$$

Question Seven

$$(a) \quad (i) \quad \cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \quad \checkmark$$

$$(ii) \quad \cos 4\theta = \frac{1 - \tan^2 2\theta}{1 + \tan^2 2\theta} \quad \checkmark$$

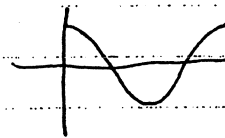
$$= \frac{1 - \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)^2}{1 + \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)^2} \quad \checkmark$$

$$= \frac{1 - 2 \tan^2 \theta + \tan^4 \theta - 4 \tan^2 \theta}{1 - 2 \tan^2 \theta + \tan^4 \theta + 4 \tan^2 \theta}$$

$$= \frac{1 - 6 \tan^2 \theta + \tan^4 \theta}{1 + 2 \tan^2 \theta + \tan^4 \theta} \quad \checkmark$$

Consider $\cos 4\theta = 0$
 $4\theta = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}$

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}$$



Consider the quadratic

$$1 - 6x + x^2 = 0 \quad \text{---} \quad *$$

where $x = \tan^2 \theta$

The solutions to * are

$$\tan^2 \frac{\pi}{8} \text{ and } \tan^2 \frac{3\pi}{8}$$

$$\Sigma \text{ roots} = \tan^2 \frac{\pi}{8} + \tan^2 \frac{3\pi}{8} = -\frac{b}{a}$$

$$\tan^2 \frac{\pi}{8} + \tan^2 \frac{3\pi}{8} = 6 \quad \checkmark$$

Question Seven (b)

(14)

$$(i) \quad \rho^9 = \rho^7 \times \rho^2 \\ = \rho^2 \quad \checkmark$$

$$(ii) \quad \alpha + \theta = \rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6$$

In the equation $z^7 - 1 = 0$ $\checkmark$
coeff of $x^6 = 0$

$$\therefore 1 + \rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6 = 0$$

$$\rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6 = -1 \quad \checkmark$$

$$(iii) \quad \alpha\theta = (\rho + \rho^2 + \rho^4)(\rho^3 + \rho^5 + \rho^6)$$

$$= \rho^4 + \rho^6 + \rho^7 + \rho^5 + \rho^7 + \rho^8 + \rho^7 + \rho^9 + \rho^{10}$$

$$= \rho + \rho^2 + \rho^3 + \rho^4 + \rho^5 + \rho^6 + \rho^7 + 2 \quad \checkmark$$

$$= 0 + 2 \quad \checkmark$$

$$= 2$$

(iv) The quadratic is

$$x^2 - (\alpha + \theta)x + \alpha\theta = 0 \quad \checkmark$$

$$x^2 + x + 2 = 0 \quad \text{---} *$$

(v) The roots of * are

$$x = \frac{-1 \pm \sqrt{7}i}{2} \quad \checkmark$$

$$\operatorname{Re}(\rho + \rho^2 + \rho^4) = -\frac{1}{2}$$

$$\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} = -\frac{1}{2} \quad \checkmark$$

3 (a) $a_1 = \frac{2!}{2^1 \times 1!} = 1$, which is odd. ✓

Suppose that $a_k = \frac{(2k)!}{2^k k!}$ is odd. ✓

Prove that a_{k+1} is odd:

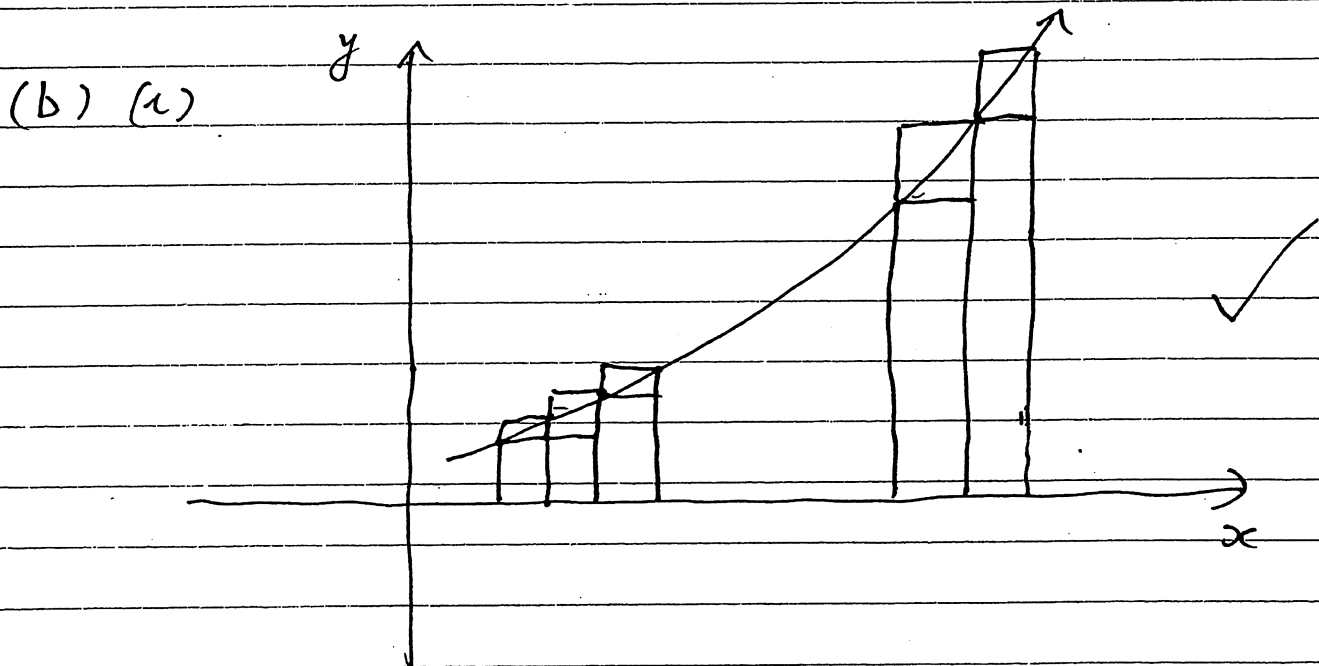
$$a_{k+1} = \frac{(2k+2)!}{2^{k+1} (k+1)!}$$

$$= \frac{(2k+2)(2k+1)(2k)!}{2 \times 2^k \times (k+1) \times k!}$$

$$= \frac{2(k+1)(2k+1)(2k)!}{2(k+1) 2^k k!}$$

$$= \underbrace{(2k+1)}_{\text{odd}} \cdot \underbrace{a_k}_{\text{odd}}$$

$\therefore a_{k+1}$ is odd ✓



sum of area
of lower rectangles

< exact area
under curve
from $x=1$ to $x=n$

< sum of areas
of upper rectangles ✓

$$f(1) + f(2) + \dots + f(n-1) < \int_1^n f(x) dx < f(2) + f(3) + \dots + f(n)$$

$$(ii) \int_1^n \ln x \frac{d}{dx}(x) dx$$

$$= [x \ln x]_1^n - \int_1^n 1 dx \quad \checkmark$$

$$= n \ln n - n + 1 \quad \checkmark$$

$$(iii) (x) \text{ Let } f(x) = \ln x$$

$$\text{From (i): } \int_1^n \ln x dx < \ln 2 + \ln 3 + \dots + \ln n$$

$$\text{Using (ii): } n \ln n - n + 1 < \ln n! \quad \checkmark$$

$$n! > e^{n \ln n - n + 1}$$

$$n! > n^n e^{1-n}$$

$$n! > \frac{n^n}{e^{n-1}} \quad \checkmark$$

$$(\beta) \text{ From (i) and (ii)}$$

$$\ln 1 + \ln 2 + \dots + \ln(n-1) < n \ln n - n + 1 \quad \checkmark$$

$$\ln(n-1)! < n \ln n - n + 1$$

$$(n-1)! < n^n \cdot e^{1-n}$$

$$n! < \frac{n^{n+1}}{e^{n-1}} \quad \checkmark$$

$$(iv) \text{ From (iii), } \frac{n^n}{e^{n-1}} < n! < \frac{n^{n+1}}{e^{n-1}}$$

$$\text{Taking } n^{\text{th}} \text{ roots: } \frac{n}{e^{1-\frac{1}{n}}} < (n!)^{\frac{1}{n}} < \frac{n^{1+\frac{1}{n}}}{e^{1-\frac{1}{n}}} \quad \checkmark$$

$$\therefore e^{\frac{1}{n}-1} < \frac{n}{\sqrt[n]{n!}} < \sqrt[n]{n} e^{\frac{1}{n}-1}$$

$$\text{As } n \rightarrow \infty \quad e^{\frac{1}{n}-1} \rightarrow e^{-1} \text{ and } \sqrt[n]{n} \rightarrow 1 \text{ given}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n} = \frac{1}{e} \quad \checkmark$$